Abhijeet Vikram

BS-MS Interdisciplinary Sciences

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Education

- **Bachelor and Master of Science (BSMS) in Interdisciplinary Sciences** - 5th year (expected 2027)
  Indian Institute of Science Education and Research, Pune, India
- **The Aryan International School, Varanasi, India**
  CBSE Class XII (2022): 93% | Class X (2020): 92%

Relevant Coursework

- Linear Algebra
- Probability
- Calculus I & II
- **Statistical Inference**
- **Markov Chain and its Application**
- Inverse Theory
- **Non-Linear Dynamics**
- **Graph Theory**
- **Stochastic Processes (auditing)**

- **Mathematical Finance**
- **Economics and Public Policy**

- Introduction to Computing
- **Statistical Learning and Data Science**
- **Deep Learning**
- **Image & Vedio Processing with Deep Learning**
- **NLP**
- **Signal Processing**

- **Neurobiology I & II**
- Bioinformatics
- **Systems Biology**
- **Ecology**
- **Chemical ecology**
- **Animal Behaviour**
- **Genome Biology**

Have worked with -
- **PINNS**
- **Random Matrix Theory**
- **Reservoir Computing**

Research Experience

- **Master's Thesis (May-Apr 2027): A study on Low-Precision Quantization and Pruning of State Space Modals for Edge Computing applications like Brain-Computer Interface (Prof. Chetan Singh Thakur - IISc Bangalore)**

- **Semester Project (Aug-Apr 2025): Diffusion model with deterministic Noising (Prof. MS Santhanam - IISER Pune)** - Report, Code

  To study diffusion model and make it energy and compute efficient **by replacing stochastic noising process with deterministic chaotic dynamics, making the reversal process faster**, making best fit for edge devices and compute constrained environments.

  It's reversal still remain a difficult problem due to loss of information and symmetric nature of logistic map in contrast to noise accumalation seen in diffusion models. This attempt still provided deep insights into Non-Linear Dynamics and Generative AI. Still working on it.

- **Summer Project (May-July 2025): Spiking Harmonic REsonate and fire SSM Model (SHaRe - SSM) (Prof. Ayon Borthakur - IIT Guwahati)**
  - We propose and rigorously evaluate a **second-order spiking SSM, completely spike-based** without using any non-linearities such as GeLU, GLU, or GSU.
  - Furthermore, we observe that SHaRE-SSM **outperforms all first-order SSMs** on **very long sequence classification** and is extremly energy efficient than ANN-based second-order SSMs.
  - Also, we extend SHaRe-SSM to energy-efficient **long-range regression** applications and propose a **convolving kernel-based regression**. We observe that SHaRe-SSM outperforms first-order SSMs on a **50K sequence length regression task**.
  - Finally, we study and explore the **impact of discretization methods** and network **heterogeneities** on SHaRe-SSM.

Publication: **NeurIPS 2025 TS4H** (Spotlight) - **A Second-Order SpikingSSM for Wearables**

We Proposed SHaRe-SSM for Wearable Devices giving SOTA performance on **Human Activity Recognition datasets and PPG-DalAi EEG datasets** while being exteremly **energy efficient (52.1x)**.

Publication: **ICML 2026** (Regular) - **A Spiking Heterogeneous Harmonic Resonate-and-Fire State Space Model for Time Series** - Webpage , Code

- **Semester Project (Jan-Apr 2025): FER2013 - ResNet (Chaitanya Guttikar - IISER Pune)**

  Worked on **FER2013 dataset** with **ResNET like architecture** and achieved an accuracy of **71.25%** by correcting the class imbalence and cleaning the dataset.
- **Winter Project (Dec-Jan 2025): Neuromorphic Computing (Prof. Ayon Borthakur - IIT Guwahati)**

  Studied **Spiking Neural Networks** and its comparison with Artificial Neural Networks and the **changes and benefits of Learnable Initial Membrane Potential and other Heterogeneities**.
- **Summer Project (Jun-Jul 2025): Evolutionary Biology (Prof. Sutirth Dey - IISER Pune)**

  Standerdized the Spatial Kernel with Drosophila Activity Monitor (DAM) and its activity data analysis. Also built an **Drosophila larvae activity tracker** to study the adult dispersal evolutionary selection pressure's effect over larval dispersal ability.
  Key Takeaways - **Data Cleaning and Data analysis using Scipy, Numpy, matplotlib, OpenCV**

- Winter Project (Dec-Jan 2023) Evolutionary Biology (Prof. Sutirth Dey)
  Standerdized few techniques (DIET, CAFE) to measure dietary intake of outbred drosophila
- Other Research Projects -
  - Effects of light weavelengths as insect attractants for pests and pollinators.
  - **Mouse training and Mouse Brain Surgery** for brain calcium imageing.
  - **PCR** to amplify TERT gene (linked to ageing and oncogenesis) with SNP and extract it.
  - Effect of Climate change on **Butterfly dispersal** - **Cox's Proportionality Hazard Model.**
  - Studied **fly carcadian rhythm** changes after crossing GMR GAL – 4 × UAS lines and their data analysis.
  - Review on **Earthquake location determination methods** - Non-Linear Methods

Skills & Interests

- **Programming:** Python (scipy, Matplotlib, Numpy, Pandas, **torch**, **Jax**), C++, Git, Latex
- **Machine Learning:** Supervised & Unsupervised Learning, Model Evaluation, Feature Engineering
- **Deep Learning:** CNN, ResNet, VGGNet, UNet, RNN, LSTM, **State Space Models (SSMs), Diffusion Models for Image generation**.
- **Interested in: Neuromorphic Computing, Spiking Neural Networks (SNNs), Low power AI systems, Mathematical Finance, PINNs, NLD and Random Matrix Theory**

Acomplishments

- **NEET (2022)**- Gen AIR 30156
- **IISER Aptitude Test (IAT)** - Gen AIR 534
- **Ramanujan Memorial Mathematics Contest** (17-18, 18-19) - CHS (BHU) Zonal Topper
- **PRMO** (Homi Bhabha Center for Science Education) (qualified for RMO), Regional Mathematics OLympiad (2019)
- **National Standard Examination in Junior Sciences** (HBCSE)

Hobbies and Achievements

- **Basketball**

  CBSE Silver Medal - Basketball under 17 Cluster
- **Hindustani Classical - Flute**

  Senior Diploma in Hindustani Classical Music Prayag Sangeet Samiti, Prayagraj, with distinction in theory